THE TRACE MAP INDUCES AN ISOMORPHISM OF GLOBAL DERIVED HOMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0B6I (sources/stacks-duality/pinned-original-source.tex line 878).

Allowed standard prerequisites (editorial).

- The sheaf level isomorphism underlying the trace map. Same chapter, Lemma labelled lemma-iso-on-RSheafHom, cited at proof lines 920-922. Supplies the isomorphism after applying DQ_Y.
- Global derived Hom as derived sections of the internal Hom. Cohomology, Section labelled cohomology-section-global-RHom, cited at proof lines 909-918. Identifies the global Homs with the sections used.
- The coherator as right adjoint to the inclusion. The adjunction of DQ_Y used at proof lines 924-931. Shows that DQ_Y isomorphisms induce isomorphisms on cohomology.

Source reference. Original equation 0B6H: sources/stacks-duality/pinned-original-source.tex, source lines 758–758. Same-chapter pointer into the author environment equation at source line 758; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0B6A: sources/stacks-chow/tags.txt, source lines 14063–14063. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 888 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9Q: sources/stacks-duality/pinned-original-source.tex, source lines 775–775. Same-chapter pointer into the author environment lemma at source line 775; invoked at proof line 899 of the counted proof. It is

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of quasi-separated and quasi-compact schemes. For all $L \in D_{QCoh}(\mathcal{O}_X)$ and $K \in D_{QCoh}(\mathcal{O}_Y)$ (0B6H) induces an isomorphism $R\mathrm{Hom}_X(L, a(K)) \to R\mathrm{Hom}_Y(Rf_*L, K)$ of global derived homs.*

Proof. By the construction in Cohomology, Section 0B6A we have

$$R\mathrm{Hom}_X(L, a(K)) = R\Gamma(X, R\mathcal{H}om_{\mathcal{O}_X}(L, a(K))) = R\Gamma(Y, Rf_*R\mathcal{H}om_{\mathcal{O}_X}(L, a(K)))$$

and

$$R\mathrm{Hom}_Y(Rf_*L, K) = R\Gamma(Y, R\mathcal{H}om_{\mathcal{O}_Y}(Rf_*L, K))$$

Thus the lemma is a consequence of Lemma 0A9Q. Namely, a map $E \to E'$ in $D(\mathcal{O}_Y)$ which induces an isomorphism $DQ_Y(E) \to DQ_Y(E')$ induces a quasi-isomorphism $R\Gamma(Y, E) \to R\Gamma(Y, E')$. Indeed we have $H^i(Y, E) = \mathrm{Ext}^i_Y(\mathcal{O}_Y, E) = \mathrm{Hom}(\mathcal{O}_Y[-i], E) = \mathrm{Hom}(\mathcal{O}_Y[-i], DQ_Y(E))$ because $\mathcal{O}_Y[-i]$ is in $D_{QCoh}(\mathcal{O}_Y)$ and DQ_Y is the right adjoint to the inclusion functor $D_{QCoh}(\mathcal{O}_Y) \to D(\mathcal{O}_Y)$. □